

GNN-ETF: A Graph Neural Network–Based Framework for Modeling Inter-Sector Relationships in ETFs

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Abstract: The integration of multiple data entities can support objectives like pattern analysis and structured representation. Since individual entities often represent aggregated collections, they inherently address specific variations. Extending this, further exploration can involve constructing models of exchange-traded funds (ETFs), leading to the concept of ETF relational modeling. Traditional optimization methods, such as performance metrics, often assume independence among assets. However, sectors are typically interconnected, and identifying as well as quantifying these links is key for ETF model construction. Graph Neural Networks (GNNs) offer a flexible approach for modeling and measuring relationships among nodes. By treating ETFs from different sectors as nodes, and their co-movement patterns at the same time point as weighted edges, we combine these associations with a performance metric to develop a GNN-ETF model, aiming to identify an optimized ETF configuration.

I. INTRODUCTION

Since there has been discussion over stocks, how to invest in them in a logical way has been a fundamental area of research that has interested financial economists, mathematicians, and other investigators. The Sharpe ratio, as a normalized performance measure of funds, itself represents the vital importance of risk in calculating portfolio performance [5]. Being a risk-adjusted measure of performance, the Sharpe ratio provides a broad metric, in a simultaneous approach, accounting for risk as well as reward, and its employment over a longer time horizon reduces the adverse impact of risk factors in performance measurement.

With progressively deeper and more mature technology of deep learning and expanded large-model applications, artificial intelligence has been applied in applying to stock investments as well. For instance, Navon et al. [6] hypothesized Financial Time Series Prediction based on Deep Learning, marking foundational work which was among earliest in employing deep learning in applying to stock investments. They proposed a data-driven, end-to-end deep learning approach to predict financial time series, trying to predict trend in stocks as well as in ETFs. Shih et al. [7] wrote Forecasting ETF Performance: A Comparative Study of Deep Learning Models and the Fama-French Three-Factor Model, where a few different models of neural network were contrasted against a classic Fama-French three-factor model, not only providing proof for how prevalent neural networks are, but its suitability in predicting stocks. Liu et al. [8] presented FinRL: A Deep Reinforcement Learning Library for Automated Stock Trading in Quantitative Finance,

where RL was applied to map to quantitative trading, providing researchers a usable toolkit that accelerates code construction and reduces distance from backtesting to end-to-end trading deployment. According to this, Li et al. [9] proposed R&D-Agent-Quant: A Multi-Agent Framework for Data-Centric Factors and Model Joint Optimization, a trailblazing work in proposing a data-centric-based multi-agent framework in quantity finance, which achieved a two-fold increase in annualized return. More recently, Duan et al. [10] proposed FactorVAE: A Probabilistic Dynamic Factor Model Based on Variational Autoencoder for Predicting Cross-Sectional Stock Returns. This work naturally integrates factor models from finance and economics together with variational autoencoders (VAEs) from deep learning, introducing a prior-posterior learning framework as a new component. Empirical experiments over real-world stock market data prove that FactorVAE outperforms several baselines, serving as an example that deep learning methodologies outside of RL as well as its popularized equivalent based on recurrent neural networks have great potential in finance applications as well. Taking inspiration from these achievements, this work attempts to bring in additional deep learning methodologies into researching the field of investment, providing researchers additional inspirations for exploration.

Graph Neural Networks (GNNs), a class of deep models applicable to learn from graph-structured data, are remarkably efficient in extracting patterns and features from graphs. GNNs have been widely applied in clustering, classification, and prediction tasks, and are specifically applied in finding associations among data entities [11]. For example, McBrearty et al. [12] applied GNNs in earthquake phase association, successfully breaking through challenges in poor seismic monitoring conditions and realizing potential practicality of GNNs. Similarly, Yang et al. [13] applied GNNs in multivariate time series classification, representing a series of potential practical uses from medicine to finance. Based on successfully verified practicability, this work applies GNNs in association analysis of ETFs, aiming to enable researchers to better understand and capitalize upon AI, and provide a new perspective for decision-makers in AI applied in financial and trading cases.

This work has several very meaningful contributions in financial modeling and portfolio optimization. First, it proposes a novel graph neural network solution for correlation

analysis of Exchange-Traded Funds (ETFs), which, to our knowledge, is one of the very earliest efforts in modeling ETFs' correlations based upon GNNs. With ETFs from different industries as graph nodes and co-movement correlations as weighted edges, such a solution provides a very robust platform for explaining cross-industry dependencies in finance. Secondly, it proposes a novel integration of the Sharpe ratio and GNN-based graph representation learning, so a hybrid scheme superiorly balancing risk and return than classical ones is facilitated. Finally, our work illuminates GNNs' real contributions toward AI-driven financial decisions, giving a very valuable tool to investors and filling a gap between advanced machine learning methods and practicable financial

strategies, broadening GNNs' applicability in portfolio construction and trading.

II. RELATED WORK

This work can be summarized as: Data collection, data preprocessing, GNN-based association analysis, data visualization, and ETF investment selection. The related research begins with data collection, during which necessary processing for studies of similar types was conducted. This approach is applicable to most studies of the same type, characterized by high versatility and robustness; details will be provided in subsequent sections (see Fig 1).

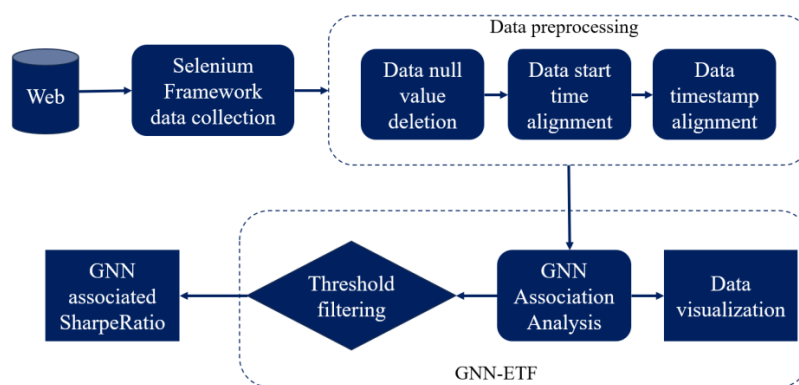


Fig. 1 Algorithm flowchart

A. Data Collection and Preprocessing

The dataset used in this study was obtained from the fund section of TongHuaShun Finance (www.0033.cn). Using browser automation, we collected data on 22 ETF sectors covering over 3,400 individual stocks from September 2017 to July 2024. Data collection was conducted via the Selenium framework, simulating real user interactions with web pages to download data. This approach ensured legal, ethical, and non-disruptive access that did not interfere with other users.

Due to the heterogeneous listing dates of ETFs, their starting times are inconsistent, while their ending times are aligned. Consequently, the total number of records for each ETF differs. In addition, online real-time scraping inevitably leads to missing records due to network fluctuations, including occasional gaps in individual ETFs for the same timestamp.

Since correlation analysis requires aligning the time horizons across ETFs, the raw dataset is not directly suitable for association analysis. Therefore, we adopted a “timestamp alignment” strategy. Specifically, we first compared the starting dates of all ETFs and selected the latest start date as the reference timeline. This improved preprocessing efficiency by approximately 31%. Second, we traversed all ETF records and stored the union of all timestamps in a set data structure. If any ETF record was missing at a particular timestamp, that timestamp was removed from the set. This method only accesses the timestamp column rather than the full dataset, thereby enhancing computational efficiency. Finally, we

extracted the aligned timestamps from the set, retrieved the corresponding records for each ETF, and generated a new aligned dataset.

After preprocessing, the final dataset contained 22 ETF sectors, each with approximately 15,800 records, totaling around 350,000 entries.

B. GNN-Based Association Analysis

Graph Neural Networks (GNNs) are specifically designed to process graph-structured data, in contrast to traditional deep learning algorithms that are primarily tailored for text (NLP) or images (CV). Due to the unique nature of graphs, they inherently encode three levels of information. First, each node contains intrinsic attributes. Second, edges between nodes encode relational information, such as connectivity and connection strength [14]. Third, the graph as a whole represents a structural information unit. For example, molecular graphs in chemistry associate structural topology with molecular properties; based on known molecular structures with established properties, one can infer potential properties of unknown molecules [15]. Overall, GNN tasks can be broadly categorized into node-level, edge-level, and graph-level analyses.

Association analysis, also referred to as association mining, aims to discover frequent patterns, correlations, and causal relationships among itemsets or objects in large-scale datasets [16]. Classical algorithms such as Apriori [17] and FP-growth [18] are widely used but exhibit limitations in efficiency when

handling large-scale data, and they lack the global representational power of GNNs. Therefore, in this study, we employ GNNs to analyze correlations among ETFs.

each ETF is treated as a node, with node features comprising historical return data and volatility metrics over the aligned timestamps. Edge definition: co-movement correlations in ETF returns, calculated using Pearson correlation coefficients based on daily returns, define the weighted edges. Graph construction: the full dataset is represented as a graph structure, where the adjacency matrix is constructed using the correlation coefficients, with a threshold of 0.3 applied to filter weak connections and reduce noise.

The GNN model is based on a Graph Convolutional Network (GCN) architecture, consisting of three graph convolutional layers. Each layer applies a graph convolution operation to aggregate information from neighboring nodes, followed by a ReLU activation function. The input node features are 64-dimensional vectors, capturing ETF returns and volatility metrics. The hidden layers have 128 and 64 units, respectively, and the output layer produces a 32-dimensional embedding for each ETF node, representing its learned representation for portfolio optimization. The model is trained using the Adam optimizer with a learning rate of 0.001, a batch size of 32, and a maximum of 200 epochs, with early stopping applied if the validation loss does not improve for 20 consecutive epochs. The loss function is a mean squared error (MSE) loss, designed to minimize the difference between predicted and actual ETF correlations. Dropout with a rate of 0.3 is applied to the hidden layers to prevent overfitting. The model is implemented using PyTorch Geometric, and training is performed on a single NVIDIA RTX 3080 GPU, with an average training time of approximately 2 hours for the dataset of 22 ETF sectors.

Following this design, the resulting GNN-based ETF association analysis is illustrated in Fig. 2.



Fig. 2 GNN-based ETF association analysis

It can be observed that ETFs in the Big Data, Communication, and Electronics sectors are highly correlated.

This implies that fluctuations in one of these ETFs are likely to affect the others. From a defensive investment perspective, it is advisable to avoid holding all three simultaneously; instead, one should select only one among them. Investors may also use this framework to assess whether their portfolios require adjustments in light of inter-sector dependencies.

C. GNN-ETF

The Sharpe ratio, also referred to as the Sharpe index or a standardized measure of fund performance, is a fundamental indicator in modern investment theory. It demonstrates that the level of risk plays a foundational role in determining portfolio performance. The risk-adjusted rate of return is a comprehensive metric that simultaneously accounts for both return and risk, effectively mitigating the adverse impact of risk factors on long-term performance evaluation. The Sharpe ratio is therefore regarded as one of the three classic indicators that comprehensively consider both risk and return.

$$\text{SharpeRatio} = \frac{E(R_p - R_f)}{\sigma_p} \quad (1)$$

where

R_p : the expected annualized return of the investment portfolio

R_f : the annualized risk-free rate

σ_p : the standard deviation of the portfolio's annualized return

As shown in Figure 3, ETF investments should not treat each ETF as an independent entity when calculating the expected annualized return of the portfolio. Instead, it is also necessary to account for holding ETFs that are highly correlated to an appropriate extent. Therefore, we integrate the GNN with the Sharpe ratio, resulting in the **GNN-ETF** model for improved investment decision-making.

$$\text{GNN-ETF} = \frac{E(R_p - R_f)}{\sigma_p + G} \quad (2)$$

Where

$$G = \sum_{k=1}^{num} \text{argmax}(abs(R_i)) * \varepsilon_k \quad (3)$$

$abs(R_i)$ aims to capture the absolute value of return fluctuations, and $\text{argmax}(abs(R_i))$ identifies the largest absolute fluctuation in returns. Based on this, we select num industries with correlations greater than a given threshold and weight them by their corresponding correlation scores ε_k .

CONCLUSION

In conclusion, the GNN-ETF framework presents a robust approach to portfolio construction by leveraging Graph Neural Networks to model co-movement relationships among ETFs, thereby enhancing investment strategies under market volatility and promoting diversification. However, the framework's current limitations include its focus on return

fluctuations without considering absolute ETF prices and its lack of explicit guidance on precise allocation or position sizes. Looking ahead, integrating reinforcement learning (RL) with the GNN-ETF framework offers promising potential to address these shortcomings by enabling dynamic strategy adjustments based on market conditions. Despite challenges posed by the large action space in financial markets, combining RL with GNN-ETF could reduce complexity through informed portfolio selection, paving the way for next-generation adaptive investment strategies.

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